# Ishan Acharya-Gangopadhyay

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## EDUCATION

**Michigan State University** — *East Lansing, MI*
B.S. Electrical Engineering — *Expected May 2029*
**Relevant Coursework:** Circuit Analysis (ECE 201), Multivariable Calculus (MTH 234), Programming in C (CSE 220)

## EXPERIENCE

**Lead Backend/Infrastructure Engineer** | ***Wensura*** — *January 2026 – Present*

- Built B2B SaaS platform for battery materials research with 5 active beta customers; led all infrastructure, security, and backend engineering
- Built AWS infrastructure from scratch with Terraform: ECS Fargate, RDS PostgreSQL + pgvector, Redis, S3/CloudFront, SQS, DynamoDB, ECR, EFS, VPC/IAM, and ALB
- Ported 3,000-line TEA monolith into a clean 5-file FastAPI package with unit-safe computation (pint + NewType hybrid), SQLAlchemy 2.0, and selective Fernet field encryption
- Engineered multi-stage LLM peer review pipeline with iterative RAG retrieval, extended-thinking gap extraction, and SSE token streaming
- Implemented PostgreSQL RLS for multi-tenant isolation via FastAPI context middleware; integrated Firebase MFA auth and Stripe billing
- Hardened CI/CD with pinned GitHub Actions SHAs, Gitleaks scanning, Checkov, and Trivy container scanning; grew test coverage from 27% → 64%

**Electrical Subteam Engineer** | ***MSU Solar Racing*** — *Fall 2025 – Present*

- Worked on LVGL-based embedded dashboard UI for solar car Lysander; implemented Xvfb headless rendering in GitHub Actions to validate UI output without physical hardware
- Sole owner of subteam CI/CD pipeline across build, test, and LVGL render validation stages

**Director of Software** | ***FRC Team 4829 Titanium Tigers*** — *2021 – 2025*

- Progressed from member to Director of Software over 4 seasons; led full-stack robot control and vision pipeline development
- Built full FRC robot digital twin in Java: dyn4j/ode4j physics arena, simulated swerve odometry with gyro + per-module state, game piece state machine with projectile dynamics, and unit-safe PID/feedforward/trapezoidal profile controllers with stator/supply current limiting throughout
- Built multithreaded pose estimation pipeline across variable camera count with automatic MegaTag1/2 switching and teleport detection; delivered team's first reliable vision system
- Introduced CI/CD and structured logging to the team for the first time; trained 10–15 members annually in Java, GitHub, robot mechanics, and control theory

## PROJECTS

**Computational Holography Simulator** | *Python* — *2025 – Present*

- Built a full CGH simulator implementing Angular Spectrum Method (ASM) propagation, Gerchberg-Saxton phase retrieval, SLM quantization modeling, and speckle noise analysis
- Successfully reconstructed complex holograms across multiple scene types and image configurations

**Self-Hosted Multi-Agent Second Brain** | *FastAPI · Ollama · Docker* — *2025 – Present*

- Designed and deployed a Docker Compose multi-agent pipeline (10+ specialized agents: Librarian, Synthesist, Researcher, Meta-Coach, etc.) orchestrated via OpenCode with MCP servers
- Integrated RAG pipeline with arXiv + Semantic Scholar retrieval; local inference on Apple M1; Telegram bot + React/Vite frontend interfaces and synced public repo

## TECHNICAL SKILLS

**Languages:** Java, Python, TypeScript, Rust, C, JavaScript
**Frameworks:** FastAPI, React, Vite, LightRAG, pgvector, LVGL
**Cloud & Infra:** AWS (ECS Fargate, RDS, S3, CloudFront, SQS, DynamoDB, ECR, EFS, VPC, IAM, Route53, Security Groups), Terraform, Docker, GitHub Actions
**Databases:** PostgreSQL, Redis, DynamoDB
**Observability:** OpenTelemetry, Grafana Cloud, Gitleaks, Checkov, Trivy
**AI / ML:** RAG pipelines, LLM orchestration, Ollama local inference, vector embeddings, SSE streaming
**Hardware:** ESP32-S3, PlatformIO, I2C/SPI, Arduino, Raspberry Pi, soldering & rework